

DEEP LEARNING INTRODUCTION

Isaac Go2 RL Training

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Agenda

1. Subject
2. Environment
3. Training & Theory
4. Result

PART 01

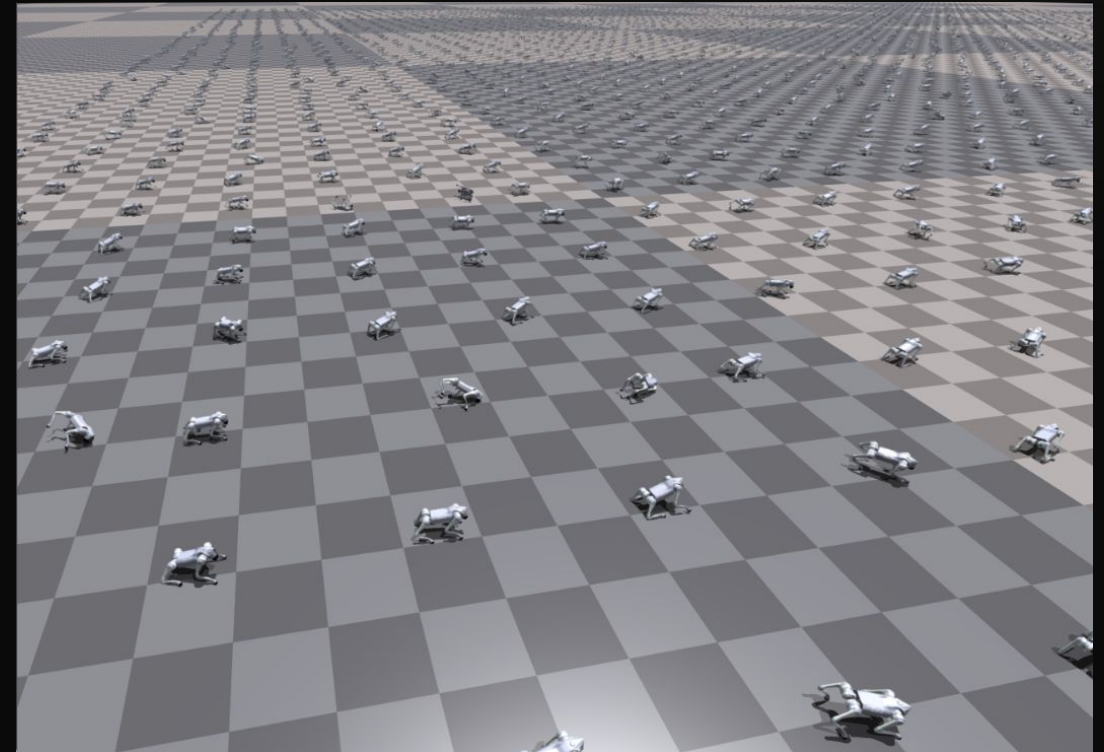
Subject

MOTIVATION

Why Deep Learning for Go2?

Learned Reflexes

Deep Learning allows the robot to learn optimal "reflexes" through trial and error, mimicking biological locomotion patterns.



MOTIVATION

What is Go2?

Complex Coordination

Unitree Go2 has 12 joints. Manual rule-based coding for stable walking on stairs or rough terrain is mathematically overwhelming.



ENVIRONMENT

Setup: Isaac Lab Platform

- **Massive Parallelism:** GPU-accelerated training with 4,096 Go2 robots simultaneously.
- **Fast Data Collection:** Simulates months of experience in just 3 hours.
- **PhysX Accuracy:** High-fidelity simulation of contact friction and joint torques.



PART 02

Training & Theory

How the Chapters Fit the Pipeline

Ch 6: Feedforward Networks

An **Actor-Critic MLP** maps proprioceptive sensor inputs (orientation, joint states) and the velocity command to 12 motor actions.

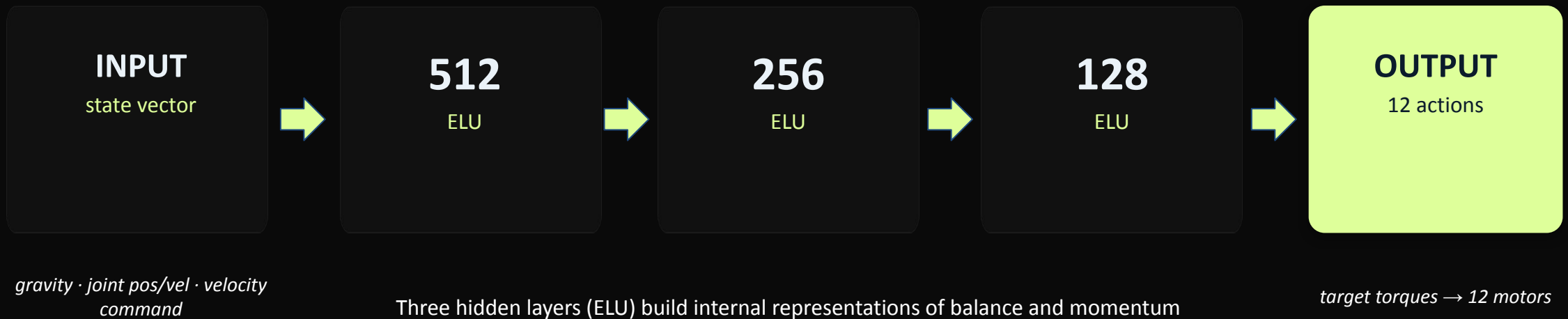
Ch 8: Optimization

PPO Algorithm ensures stable weight updates. It clips the policy update ratio to prevent the robot from "forgetting" how to stand.

Ch 7: Regularization

Early Stopping saves the highest-reward weights as `best_agent.pt`, so the robot keeps its best gait even if later training degrades.

The Policy Network



Why deep learning here: a 12-DoF quadruped is a non-linear control problem — the MLP learns the mapping that explicit equations can't tractably express.

Optimization at Scale

PPO surrogate loss

Optimize a policy-gradient objective that maximizes reward and penalizes falling — not a simple MSE.

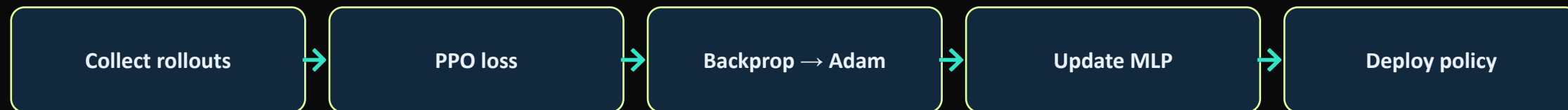
Adam optimizer

Per-weight adaptive learning rates from gradient moments — essential for this loss landscape.

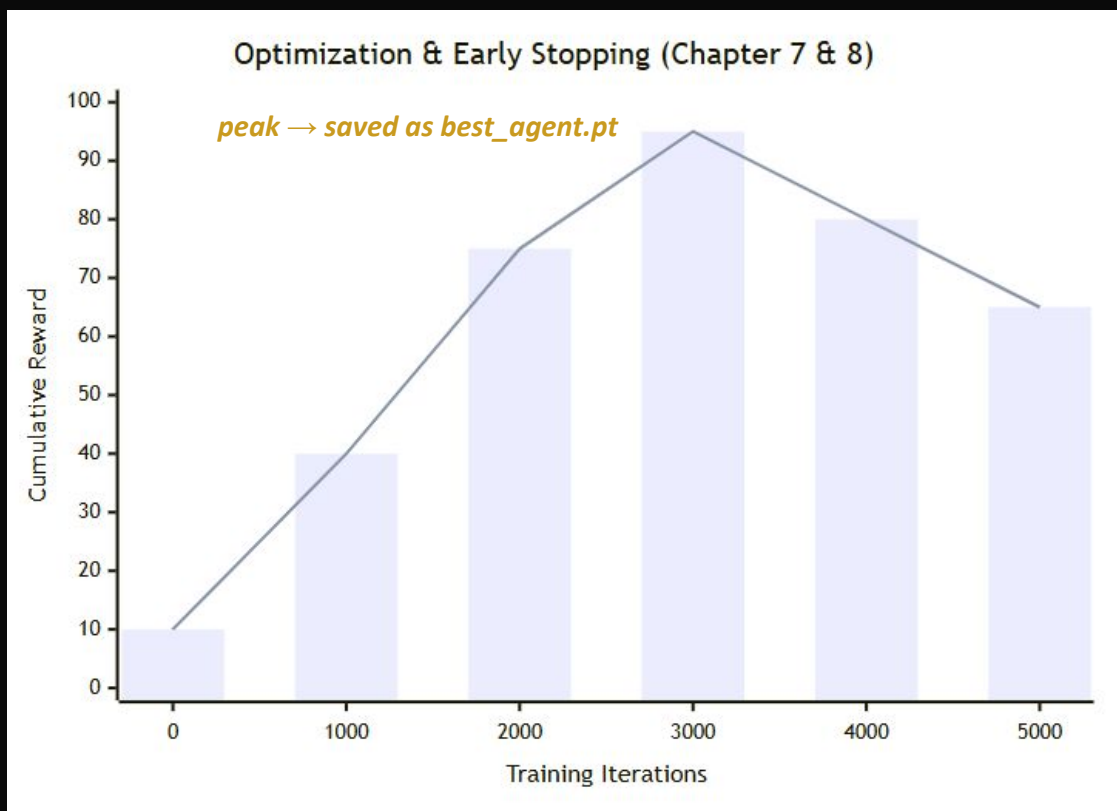
4,096 parallel envs

Massive mini-batches collected in Isaac Sim cut RL variance and stabilize each gradient step.

THE TRAINING LOOP



Regularization



The instability

RL training can degrade — a policy that walks at iteration 3,000 may over-explore and “forget” by 4,000, like overfitting to recent batches.

The fix: Early Stopping

SKRL tracks average episodic reward and isolates the highest-scoring weights as *best_agent.pt* — a regularizer that always keeps the most generalized policy.

PART 03

The Final Results

APPLICATION

Hierarchical Control

The learned DL policy acts as a "low-level reflex" that enables:

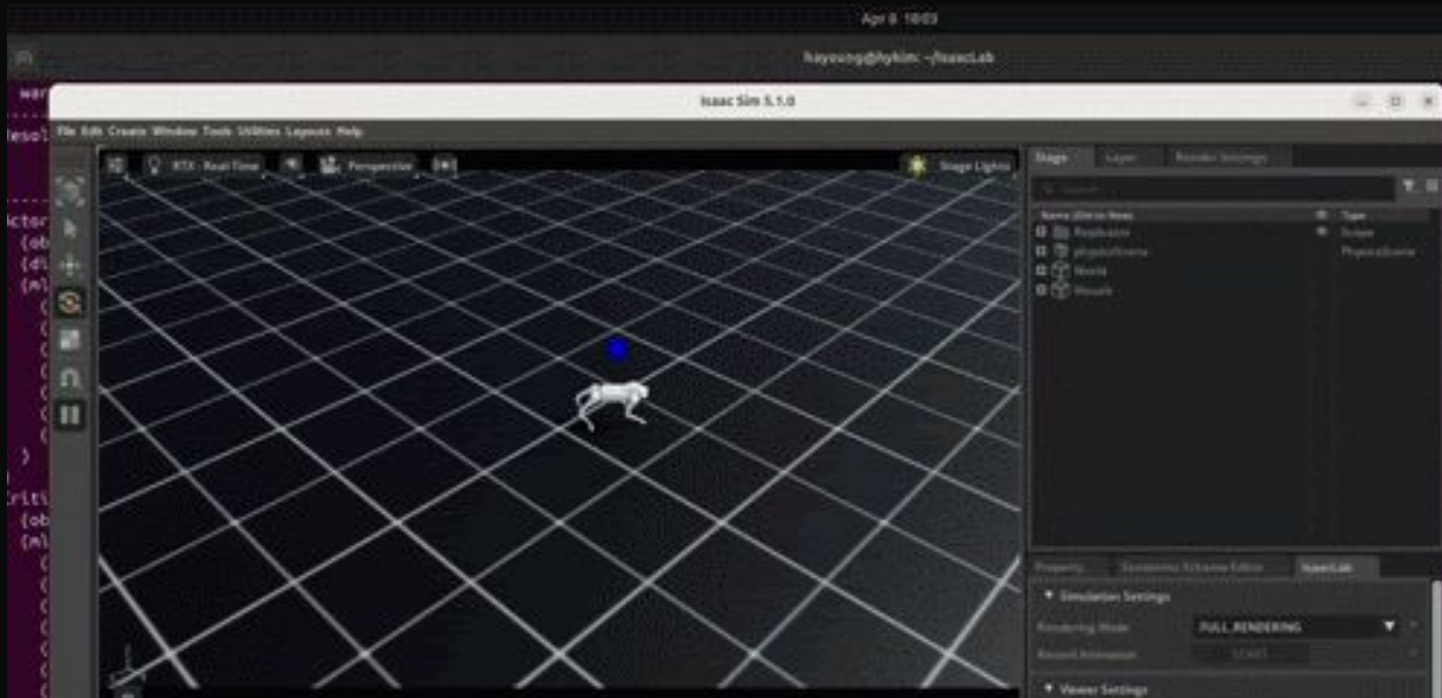
- **SLAM**: Mapping unknown rooms.
- **Nav2**: Calculating paths.
- **RL Policy**: Precise foot placement.

The DL model executes high-level commands flawlessly on difficult terrain.



STEP 1.

Isaac-lab - RL policy



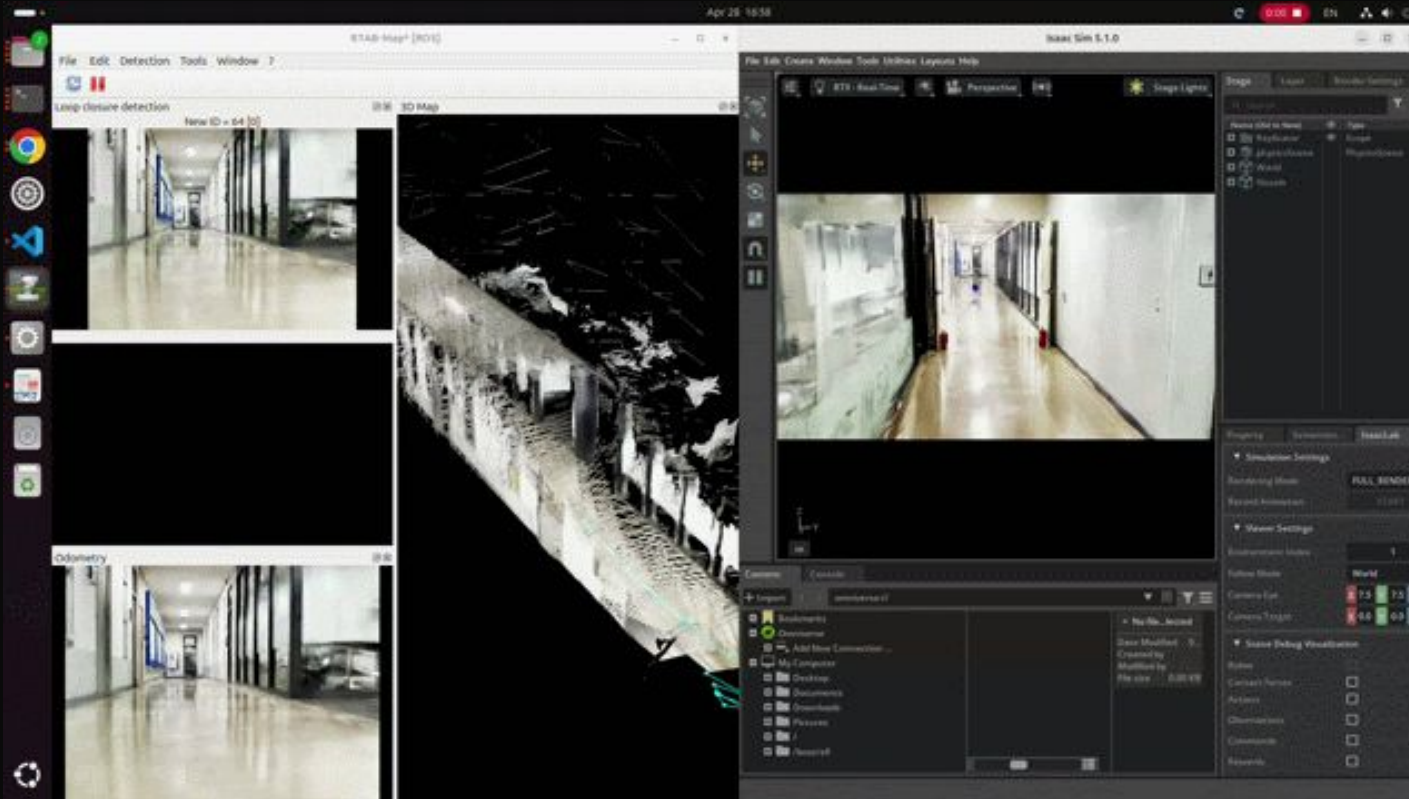
Early robot stage

insufficient training example.

Robot falls easily due to limited learning.

STEP 2.

SLAM - RTAB-MAP



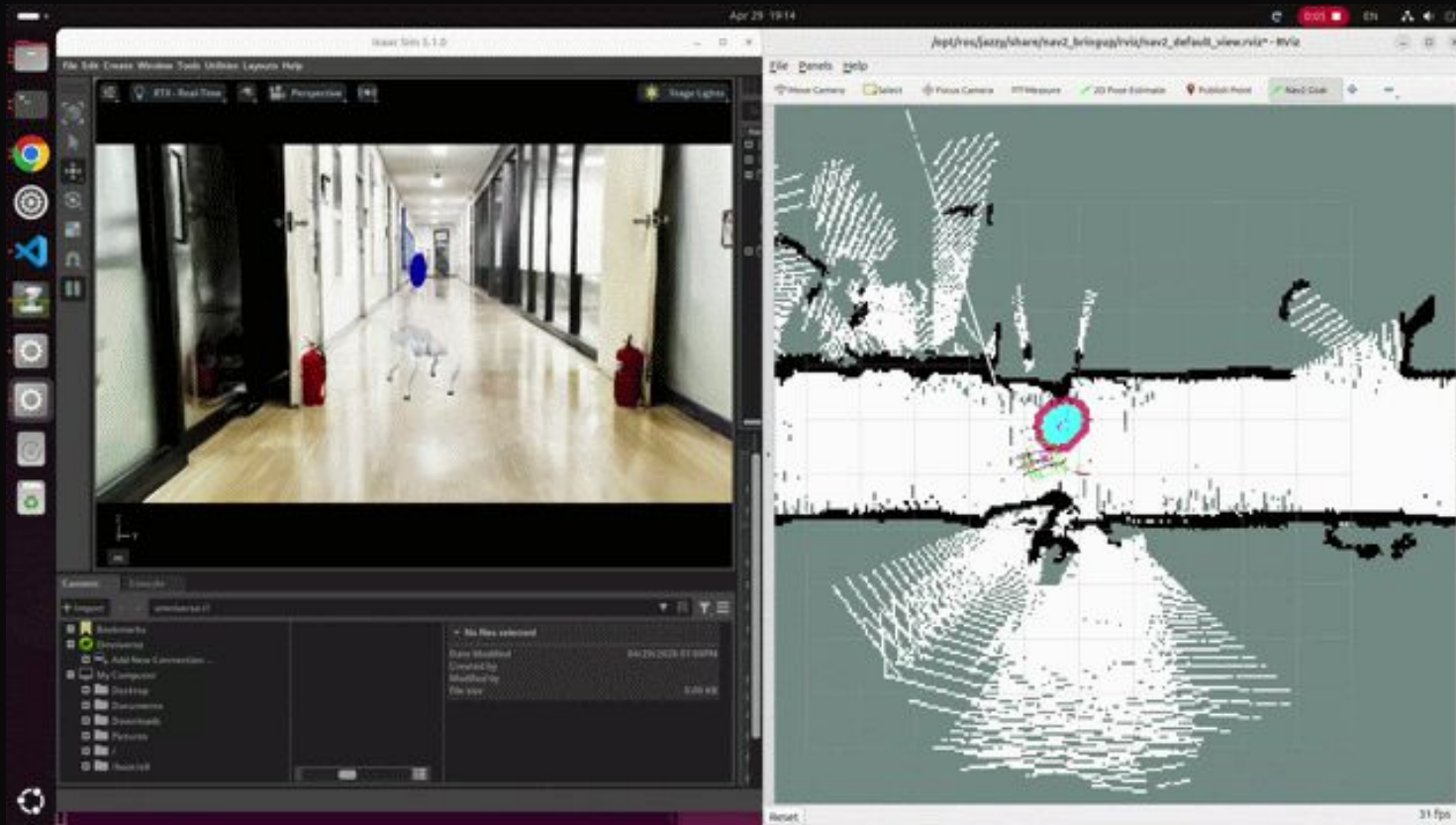
VisualSlam - mapping

Well-trained robot navigates and maps the space.

Keyboard-controlled driving with simultaneous mapping.

STEP 3.

Nav2 - Autonomous driving



Localization & driving

Robot autonomously navigates to target coordinates.

Autonomous navigation using Nav2
with 2D map guidance

Robot locomotion validated through
sensor-controlled walking

STEP 4.

What's next? - SIM To Real



Bridging the gap between simulation and reality.

Deep learning enables real-world application.
Reducing Sim-to-Real gap through deep learning.

Thank *you!*
